PRACTICE SHEET-1

1. If n(A) = 115, n(B) = 326, n(A – B) = 47, then what is n(A∪B) equal to?
   (1) 373 (2) 165
   (3) 370 (4) 394

2. If P(A) denotes the power set of A and A is the void set, then what is number of elements in P{P{P{P(A)}}}?
   (1) 0 (2) 1
   (3) 4 (4) 16

3. If $N_a = \{ax \mid x \in N\}$, then what is $N_{12} \cap N_8$ equal to?
   (1) N_{12} (2) N_{20}
   (3) N_{24} (4) N_{48}

4. If $X = \{(4^n - 3n - 1) \mid n \in N\}$ and $Y = \{9(n - 1) \mid n \in N\}$, then $X \cup Y$ equals to
   (1) X (2) Y
   (3) N (4) A null set

5. What is the number of proper subsets of a given finite set with n elements?

11. If A, B and C are subsets of a given set, then which one of the following is relations is
    (a) A∪ (A∩B) = A∪B
    (b) A∩(A∪B) = A
    (c) (A∩B)∪C = (A∪C)∩(B∪C)
    (d) (A∪B)∩C = (A∩C)∪(B∩C)

12. If a set A contains 3 elements and another set B contains 6 elements, then what is the minimum number of elements that (A∪B) can have?
    (a) 3 (b) 6
    (c) 8 (d) 9

13. If A = {x: 0 ≤ x ≤ 2} and B = {y: y is a prime number}, then what is A∩B equal to?
    (a) ∅ (b) {1}
    (c) {2} (d) {1,2}

14. If A = {x : x is a multiple of 2}, B = {x : x is a multiple of 5} and C = {x : x is a multiple of 10}, then A ∩ {B ∩ C} is equal to:
    (a) A